

Generative Models in Finance

Week 3: LLMs in Finance and Investment Management

Cristopher Salvi
Imperial College London

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Overview

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Reference: Y. Kong, Y. Nie, X. Dong, J. M. Mulvey, H. V. Poor, Q. Wen & S. Zohren, *Large Language Models for Financial and Investment Management: Applications and Benchmarks*, Journal of Portfolio Management, 2024.

Reading Between the Lines

“Recent indicators suggest that economic activity has continued to expand at a solid pace. Job gains have slowed, and the unemployment rate has moved up but remains low. Inflation has made further progress toward the Committee’s 2 percent objective but remains somewhat elevated.”

— FOMC Statement, September 2024

- This paragraph moved Treasury yields within minutes of release
- The language is deliberately calibrated: “solid pace” signals confidence, “somewhat elevated” signals caution. The precise choice of words *is* the message
- A macro strategist reads these nuances instantly — but cannot read 10,000 documents per day
- **Question:** can a machine extract the same signal from text, at scale?

The Scale of the Problem

- Quantitative finance has traditionally relied on **structured data**: prices, volumes, balance sheet items — numbers in rows and columns
- But the vast majority of financial information is **unstructured text**:
 - ▶ 10-K filings: $\sim 50,000$ words each $\times \sim 4,000$ US public companies
 - ▶ Earnings calls: $\sim 3,000$ per quarter, each 30–60 minutes
 - ▶ News: thousands of articles per day from Reuters, Bloomberg, the FT
 - ▶ Social media: millions of daily posts on Twitter/X and StockTwits
 - ▶ Central bank communications: FOMC minutes, ECB statements, speeches
- A single equity analyst covers ~ 15 stocks. The gap between available text and human capacity is enormous
- **LLMs are the first technology that can process unstructured text at this scale** and extract structured, quantitative signals. The rest of this lecture examines how — and where they fall short

What Is Sentiment Analysis?

- **Sentiment analysis** assigns a score (positive, negative, or neutral) to a piece of text, quantifying its opinion or tone
- **Financial application:** given a news headline, an earnings transcript, or a central bank statement, estimate how *positive* or *negative* the market-relevant content is
- **Why it matters:** if sentiment can be measured *before* the market fully reacts, it becomes a trading signal
- **Core sub-tasks** in the literature:
 - ▶ Scoring news headlines and social media posts
 - ▶ Analysing tone in corporate disclosures and earnings calls
 - ▶ Interpreting central bank communications (FOMC minutes, ECB statements)

From Word Counts to Contextual Understanding

- **Layer 1 — Dictionaries.** Count positive/negative words from predefined lists. The Loughran–McDonald (2011) dictionary is finance-specific: “liability” is negative in everyday English but neutral in a filing. Simple and interpretable, but blind to negation and context (e.g. “no significant decline in margins” is flagged negative because of “decline”)
- **Layer 2 — Static embeddings.** Word2Vec, GloVe: represent words as vectors in $\mathbb{R}^d$. Capture semantic similarity, but produce the same vector for “bank” whether it means a financial institution or a river bank
- **Layer 3 — Contextualised models** (BERT onward): every token’s representation depends on its full surrounding context. The model distinguishes word senses and recognises negation. This is the key transition: from counting words to understanding sentences

Finance-Specific Language Models

- **FinBERT** (Araci 2019): BERT (110M parameters) fine-tuned on financial news. Learns domain-specific word senses. The most widely used baseline for financial sentiment
- **BloombergGPT** (Wu et al. 2023): 50 billion parameters, pre-trained from scratch on 363B tokens of Bloomberg's proprietary financial text. Outperforms general models on financial NLP while remaining competitive on general benchmarks
- **General-purpose models** (GPT-4, Claude, LLaMA): not finance-specific, but often competitive with or superior to fine-tuned models, especially on tasks requiring broad world knowledge
- **A note on terminology**: the literature uses “LLM” broadly, from BERT (110M params) to GPT-4 (>1T). The capability gap is substantial — keep this in mind when interpreting results

Does LLM Sentiment Predict Returns?

- **Lopez-Lira & Tang (2023)**: GPT-4 scores a single news headline → predicts the *next-day* cross-sectional stock return. A daily long/short strategy achieves a Sharpe ratio of ~ 3.8 . Effect strongest for **small caps** and after **negative news** — consistent with slow information diffusion
- **Chen, Kelly & Xiu (2022)**: use BERT/RoBERTa embeddings of news articles to form daily-rebalanced portfolios → annualised Sharpe ratio of ~ 4.8 (2004–2019), significantly above Word2Vec baselines. Prices respond to news with a **multi-day delay**, creating a short-term trading window
- **Caveat**: both studies report in-sample or simulated results with high turnover. Transaction costs, slippage, and out-of-sample decay can erode these Sharpe ratios substantially

How Fragile Is the Signal?

- **Adversarial fragility:** Leippold (2023) replaces a few words with synonyms (“revenue grew” → “revenue expanded”) and re-queries GPT-3. These minimal, meaning-preserving edits **flip the sentiment score**
- The Loughran–McDonald dictionary is *immune* to this: synonyms map to the same polarity. LLM sophistication introduces a new attack surface that simpler methods do not have
- **Strategic adaptation:** Cao et al. (2023) document that corporations are already adjusting the language of their disclosures, knowing that AI systems read them. This is a new form of strategic communication aimed at *machine* readers — a feedback loop between AI and corporate disclosure
- **Implication:** if a competitor or bad actor can manipulate the text inputs to your LLM-based strategy (crafted press releases, social media posts), the resulting trading signals can be systematically corrupted

Sentiment: Limits and Verdict

- **Multimodal gaps:** Curti & Kazinnik (2023) find that *nonverbal* cues in FOMC press conferences (facial expressions, tone of voice) carry significant information about monetary policy. Text-only LLMs miss this entirely
- **Social media noise:** Twitter/X and StockTwits contain misinformation, sarcasm, and coordinated manipulation campaigns. LLMs may not distinguish genuine sentiment from deliberate deception
- **Out-of-sample fragility:** most studies test on 1–3 years of US equity data. Persistence across market regimes, asset classes, and geographies remains largely untested
- **Signal decay:** if many participants deploy similar LLM sentiment strategies, the arbitrage opportunity should shrink as markets incorporate information faster
- **Verdict:** LLM sentiment is a promising *input* to a broader quantitative model, not a standalone trading strategy. Robustness testing — across time, assets, and adversarial conditions — is essential before deployment

Can Language Models Understand Numbers?

- LLMs are trained on *text*. A financial time series is a sequence of *numbers*. At first glance, these have nothing in common
- But both are **sequential data**: a sentence is a sequence of tokens, a price series is a sequence of values. The transformer architecture processes sequences — it does not inherently care whether the tokens represent words or digits
- **Key insight**: the web text that LLMs are pre-trained on contains vast amounts of numerical sequences — tables of stock prices, macroeconomic data in Wikipedia, financial statements in HTML. The model has already seen numbers in temporal context

Zero-Shot Forecasting with LLMs

- **Key idea** (Gruver et al. 2023): take a pre-trained LLM (GPT-3, LLaMA-2) with *no* time-series training. Convert a numerical series into a text string — e.g. "1423.5, 1418.2, 1425.7, ..." — and prompt the model to continue the sequence
- **Why does this work at all?** LLMs are trained on vast text corpora that include tables, financial reports, and numerical data. They learn implicit patterns: trends, seasonality, and repetition. Their ability to represent *multimodal distributions* over next tokens lets them capture uncertainty naturally
- **Result:** on standard benchmarks (Darts, Monash), zero-shot GPT-3 matches or exceeds purpose-built models trained on each dataset. This is surprising — the model was never explicitly trained to forecast
- **Limitations:** (i) tokenisation splits numbers into character fragments, so the model has no native sense of numerical magnitude; (ii) RLHF-aligned models (e.g. GPT-4) perform *worse* than base models, likely because alignment distorts the output distribution; (iii) performance degrades on noisy, low signal-to-noise series typical of finance

Time-Series Foundation Models

- Gruver et al. showed that LLMs can forecast zero-shot, but repurposing a *language* model for *numerical* data is inefficient. This has motivated **purpose-built foundation models** for time series: pre-train a transformer once on large-scale time-series data, then apply it zero-shot to new forecasting problems
- **TimesFM** (Das et al. 2024, Google): decoder-only transformer pre-trained on >100B real-world time points. Produces *point forecasts*. Zero-shot performance competitive with supervised per-dataset models on standard benchmarks
- **Chronos** (Ansari et al. 2024, Amazon): repurposes the T5 (encoder–decoder) architecture. Tokenises continuous values via scaling and uniform quantisation into discrete bins, turning forecasting into a language-modelling problem over bin indices
- **Moirai** (Woo et al. 2024, Salesforce): encoder-only transformer that outputs *probabilistic* forecasts via mixture distributions — providing prediction intervals, not just point estimates. Handles multivariate input natively
- **Open question for finance**: these models are evaluated primarily on smooth, high signal-to-noise benchmarks (weather, electricity, traffic). Financial time series are noisier, non-stationary, and adversarial — performance on financial data remains largely untested

Das, A. et al. (2024). A Decoder-Only Foundation Model for Time-Series Forecasting. *ICML* 2024.
Ansari, A.F. et al. (2024). Chronos: Learning the Language of Time Series. arXiv:2403.07815.
Woo, G. et al. (2024). Unified Training of Universal Time Series Forecasting Transformers. *ICML* 2024.

Reality Check: LLMs vs. Traditional Models

- **Xie et al. (2023)**: systematic zero-shot evaluation — ChatGPT **underperforms** both traditional ML (gradient-boosted trees, SVMs) and state-of-the-art deep learning on stock movement prediction
- **Why?** Two fundamental problems:
 - 1 **Numerical precision**: tokenising “1423.50” as a string of characters loses magnitude — the model has no native understanding that 1423 is close to 1400 and far from 14
 - 2 **Distributional shift**: financial series are non-stationary. An LLM pre-trained on historical text has no mechanism to adapt to regimes it has never seen
- **The efficient market objection**: if a publicly available LLM can predict returns from publicly available news, the signal should be arbitrated away rapidly. Persistent predictability would imply a market inefficiency hard to reconcile with decades of asset pricing research
- Even the foundation models (TimesFM, Chronos, Moirai) are primarily evaluated on *general* benchmarks (weather, electricity, traffic). Financial series — noisier, more adversarial, lower signal-to-noise — are less tested

Time Series: Pitfalls and Verdict

- **Look-ahead contamination:** many studies use LLMs trained *after* the test period. The model may have memorised events or patterns from its training data that overlap with the backtest window — a critical methodological flaw that is easy to miss and hard to rule out
- **Reproducibility:** commercial APIs (GPT-3.5, GPT-4) change silently over time. A study run in January 2023 may not reproduce in June 2023. Rigorous out-of-sample testing is difficult when the model is a black box that updates without notice
- **Verdict:** LLMs are most convincing as **feature extractors** — generating text-derived signals (sentiment, entity relationships, summaries) that feed into classical quantitative models (regression, portfolio optimisation). Using an LLM as a direct price forecaster remains speculative and under-evidenced

What Is an LLM Agent?

- A standard LLM is **reactive**: you give it a prompt, it returns a response, the interaction ends
- An **LLM agent** adds three capabilities that turn it into an autonomous system:
 - 1 **Tool use**: the model can call external functions — fetch a stock price, query a database, execute a trade via an API, run Python code
 - 2 **Planning**: the model breaks a complex goal into sub-steps and executes them sequentially, deciding at each step what to do next
 - 3 **Memory**: the model maintains persistent state across interactions — it remembers past actions and outcomes
- The agent operates in a **loop**: perceive the environment → reason → act → observe the result → repeat
- **In finance**: read today's market data and news → form a trading thesis → place an order → observe the P&L → update the strategy

From Simple Bots to Autonomous Traders

- The simplest trading agent: prompt an LLM with today's news, ask "buy or sell?".
Problem: no memory — each query starts from scratch, no learning from past trades
- **FINMEM** (Yu et al. 2024): working memory + layered long-term memory (past trades, outcomes, lessons) prioritised by relevance and recency. The agent evolves its strategy through continuous learning
- **FinAgent** (Zhang, Zhao et al. 2024): the first *multimodal* trading agent — text, numerical data, and candlestick chart images via GPT-4V. Outperforms FINMEM by adding visual pattern recognition
- **QuantAgent** (Wang, Yuan et al. 2024): self-improving agent with two loops — an inner loop refines strategies via a knowledge base, an outer loop validates on real market data. Discovers trading signals autonomously
- **Alpha-GPT 2.0** (Yuan et al. 2024): human-in-the-loop — a trader proposes alpha ideas, the agent formalises, backtests, and refines them iteratively

Yu, Y.H. et al. (2024). FinMem. *AAAI Symposium Series*, Vol. 3, pp. 595–597.

Zhang, W. et al. (2024). FinAgent. arXiv:2402.18485.

Wang, S. et al. (2024). QuantAgent. arXiv:2402.03755. Yuan, H. et al. (2024). Alpha-GPT 2.0. arXiv:2402.09746.

Multi-Agent Systems and Market Simulation

- **TradingAgents** (Xiao et al. 2024): mimics the structure of a real trading firm — specialised LLM agents act as fundamental analysts, sentiment analysts, technical analysts, and traders. Agents debate and synthesise reports before the trader decides
- **StockAgent** (Zhang, Liu et al. 2024): populates a simulated market with LLM agents. Removing information channels (bulletin boards) significantly alters price dynamics — revealing how information flow drives price formation. This connects directly to market microstructure theory
- **“Homo Silicus”** (Horton 2023): LLM agents endowed with behavioural-economics preferences simulate human economic decisions. These “silicon subjects” can replace costly human experiments in social science
- **No strong evidence** that any LLM agent consistently generates alpha in *live* markets. All results are simulated or backtested
- **Verdict**: multi-agent systems are powerful for *research* (understanding market dynamics) and *augmentation* (assisting human traders). Fully autonomous LLM trading remains speculative

Hallucination and Bias

- **Hallucination:** LLMs generate text that is fluent, confident, and factually wrong — they optimise for plausibility, not truth. In finance, a hallucinated revenue figure or a fabricated analyst rating is indistinguishable from a real one. Unlike a search engine that returns “no results,” an LLM *always* produces an answer
- **Mitigation:** Retrieval-Augmented Generation (ground answers in source documents), code-based reasoning (compute rather than generate numbers), multi-agent verification
- **Bias:** Chuang & Yang (2022) find BERT and FinBERT systematically prefer certain stocks (e.g. Tesla) over others (e.g. Ford) in fill-in-the-blank prompts. An investment system built on these embeddings inherits these biases as systematic tilts
- **Fairness:** Lakkaraju et al. (2023) ask ChatGPT and Bard identical credit-card recommendation questions phrased in standard English, African American Vernacular English (AAVE), and Telugu. The same financial scenario yields **different product recommendations** depending on dialect — a user writing in AAVE may receive worse advice than one writing in standard English, despite identical financial circumstances
- **Over-reliance:** Niszczoła & Abbas (2023) show that GPT-4 scores 99% on standard financial literacy tests (vs. 65% for GPT-3.5). But in a savings-dilemma experiment, participants adopted GPT's advice heavily (weight-on-advice = 0.65), rising to 0.74 for users with *low* subjective financial knowledge — precisely the group most vulnerable to the model's remaining 1% of errors

Chuang, C. & Yang, Y. (2022). Buy Tesla, Sell Ford. *ACL*, pp. 100–105.

Lakkaraju, K. et al. (2023). LLMs for Financial Advisement. *ACM AI in Finance*, pp. 100–107.

Niszczoła, P. & Abbas, S. (2023). GPT Has Become Financially Literate. *Finance Research Letters* 58: 104253.

Sarmah, B. et al. (2023). Towards Reducing Hallucination in Extracting Information from Financial Reports. *Int. Conf. on AI-ML Systems*.

Regulation and Liability

- **The core question:** if an LLM gives financial advice and the client loses money, who is liable — the LLM provider, the firm that deployed it, or the individual who acted on it?
- **Existing frameworks were not designed for AI:**
 - ▶ *Fiduciary duty* requires advisors to act in the client's best interest — but an LLM has no concept of “interest”
 - ▶ *Suitability requirements* mandate that advice match the client's risk profile — but an LLM may not reliably distinguish a retiree from a day trader
 - ▶ *Explainability* (e.g. MiFID II): firms must explain *why* a recommendation was made — chain-of-thought reasoning may or may not satisfy this standard
- Cao et al. (2023) document that corporations are adapting disclosures to be more “machine-readable” — a feedback loop between AI systems and corporate communication
- **No jurisdiction has yet established clear rules** for AI-generated financial advice. This regulatory vacuum is a risk for both firms and consumers

Three Takeaways

- ① **LLMs unlock unstructured financial text at scale — this is real and valuable.** The vast majority of financial information was previously inaccessible to systematic strategies. LLMs make it analysable for the first time
- ② **They are best used as feature extractors, not standalone forecasters.** The strongest evidence supports LLM-derived sentiment as an *input* to portfolio construction. Using an LLM as a direct price predictor or autonomous trader remains speculative
- ③ **Robustness is the binding constraint on deployment.** Adversarial fragility, out-of-sample instability, hallucination, bias, and regulatory uncertainty all limit real-world use. **Hybrid systems** — LLM reasoning combined with classical risk models and rigorous statistical testing — are the most promising path forward